

Papers on Compiler Optimizations: Analysis and Transformations (1952-1994)

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Title	Venue Type	Venue Name	Type	Year	Sources
The problem of simplifying truth functions [9]	T&F	AMM	J	1952	O
Minimization of Boolean functions [10]	Bell Labs	Bell System Tech. J.	J	1956	O
An algorithm for translating Boolean expressions [11]	ACM	JACM	J	1962	S
High speed compilation of efficient object code [12]	ACM	CACM	J	1965	D
Peephole optimization [13]	ACM	CACM	J	1965	E
Program optimization [14]	-	PP	C	1966	A, D, E, M, O, S
Index Register Allocation [15]	ACM	JACM	J	1966	E
Analysis of Programs for Parallel Processing [16]	IEEE	TC (TEC)	J	1966	O
Object code optimization [17]	ACM	CACM	J	1969	A, D, E
Expression optimization using unary complement operators [18]	ACM	PLDI (SCO)	C	1970	E
Local optimizations [19]	ACM	PLDI (SCO)	C	1970	E
Detection and parallel execution of independent instructions [20]	IEEE	TCO	J	1970	S
Global common subexpression elimination [21]	ACM	PLDI (SCO)	C	1970	D, E, M
The Generation of Optimal Code for Arithmetic Expressions [22]	ACM	JACM	J	1970	D, E, M
Control flow analysis [23]	ACM	PLDI (SCO)	C	1970	D, E
A Basis for Program Optimization [24]	IFIP	NH	C	1971	A, D, E, M, O, S
A catalogue of optimizing transformations [25]	-	PH	C	1972	A, D, E, M, O, S
Flow graph reducibility [26]	ACM	STOC	C	1972	D
Use-definition chains with applications [27]	Elsevier	COLA (COMLAN)	C	1972	E
A global flow analysis algorithm [28]	T&F	JCM	J	1972	E
Safety of code motion [29]	T&F	JCM	J	1972	O
On the Number of Operations Simultaneously Executable in Fortran-Like Programs and Their Resulting Speedup [30]	IEEE	TCO	J	1972	D, O
Smallest-last ordering and clustering and graph coloring algorithms [31]	ACM	JACM	J	1973	-
Complete register allocation problems [32]	ACM	STOC	C	1973	
Testing flow graph reducibility [33]	ACM	STOC	C	1973	A
A unified approach to global program optimization [34]	ACM	POPL	C	1973	A, D, E, M, O, S
Fast algorithms for the elimination of common subexpressions [35]	Springer	Acta Inf.	J	1973	A, D, E, O

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Title	Venue Type	Venue Name	Type	Year	Sources
Interprocedural Analysis and the Information derived by it [36]	Springer	Prog. Meth.	J	1974	-
Register allocation via usage counts [37]	ACM	CACM	J	1974	A
Analysis of structured programs [38]	ACM	STOC	C	1974	D
Characterizations of Reducible Flow Graphs [39]	ACM	JACM	J	1974	D, E
The parallel execution of DO loops [40]	ACM	CACM	J	1974	D, O
Interprocedural Data Flow Analysis [41]	-	IFIP	J	1974	D, O
Application of lattice algebra to loop optimization [42]	ACM	POPL	C	1975	-
Program optimization - theory and practice [43]	ACM	PLDI (SCC)	C	1975	-
Time and parallel processor bounds for linear recurrence systems [44]	IEEE	TCO	J	1975	S
A simple algorithm for global data flow analysis problems [45]	SIAM	SICOMP	J	1975	A, E, O
Program improvement by source to source transformation [46]	ACM	POPL	C	1976	-
A program data flow analysis procedure [47]	ACM	CACM	J	1976	A
Recursion analysis for compiler optimization [48]	ACM	CACM	J	1976	O
Optimal Code Generation for Expression Trees [49]	ACM	JACM	J	1976	A, D, E
A Fast and Usually Linear Algorithm for Global Flow Analysis [50]	ACM	JACM	J	1976	E, O
Code generation for expressions with common subexpressions [51]	ACM	POPL	C	1976	E, S
Global data flow analysis and iterative algorithms [52]	ACM	JACM	J	1976	E, O
Compiler Analysis of the Value Ranges for Variables [53]	IEEE	TSE	J	1977	-
On Live-Dead Analysis for Global Data Flow Problems [54]	ACM	JACM	J	1977	A
Symbolic evaluation and the global value graph [55]	ACM	POPL	C	1977	A
High-level data flow analysis [56]	ACM	CACM	J	1977	A
Abstract interpretation [57]	ACM	POPL	C	1977	D
An algorithm for reduction of operator strength [58]	ACM	CACM	J	1977	E
A transformation system for developing recursive programs [59]	ACM	JACM	J	1977	S
Arithmetic shifting considered harmful [60]	ACM	SIGPLAN Notices	J	1977	S
Monotone data flow analysis frameworks [61]	Springer	Acta Inf.	J	1977	A, D, E, O
Program Improvement by Source-to-Source Transformation [62]	ACM	CACM	J	1977	D, O
An analysis of inline substitution for a structured programming language [63]	ACM	CACM	J	1977	M, S
Program Optimization Using Invariants [64]	IEEE	TSE	J	1978	-
A new method for compiler code generation [65]	ACM	POPL	C	1978	D
A practical interprocedural data flow analysis algorithm [66]	ACM	CACM	J	1978	A, D, E, O
Data Flow Analysis for Procedural Languages [67]	ACM	JACM	J	1979	A
Constructing the Call Graph of a Program [68]	IEEE	TSE	J	1979	O
Data flow languages [69]	IEEE	MARK	W	1979	S
Time and parallel processor bounds for Fortran-like loops [70]	IEEE	TCO	J	1979	S
Unrolling loops in Fortran [71]	Wiley	SPE	J	1979	S
A fast algorithm for finding dominators in a flowgraph [72]	ACM	TOPLAS	J	1979	A, E, M, O
An efficient way to find the side effects of procedural calls and the aliases of variables [73]	ACM	POPL	C	1979	A, D, E, S
Global optimization by suppression of partial redundancies [74]	ACM	CAMC	J	1979	A, D, E, O, S
Predicting the effects of optimization on a procedure body [75]	ACM	PLDI (SCC)	C	1979	E, S
Structural analysis: A new approach to flow analysis in optimizing compilers [76]	Elsevier	COLA (COMLAN)	C	1980	A
The design and application of a retargetable peephole optimizer [77]	ACM	TOPLAS	J	1980	E
Data flow supercomputers [78]	IEEE	Computer	J	1980	S
Program optimization and parallelization using idioms [79]	ACM	POPL	C	1980	-
High-speed multiprocessors and compilation techniques [80]	IEEE	TCO	J	1980	S
A composite algorithm for strength reduction and code movement optimization [81]	Springer	ACIS	J	1980	E

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Title	Venue Type	Venue Name	Type	Year	Sources
Interprocedural data flow analysis in the presence of pointers, procedure variables, and label variables [82]	ACM	POPL	C	1980	A, E, O
Deciding Linear Inequalities by Computing Loop Residues [83]	ACM	JACM	J	1981	D
A precise inter-procedural data flow algorithm [84]	ACM	POPL	C	1981	O, S
Register allocation via coloring [85]	Elsevier	COLA (COMLAN)	C	1981	A, D, E, O, S
Reduction of operator strength [86]	NJ	Program flow analysis	J	1981	A, E, S
Dependence graphs and compiler optimizations [87]	ACM	POPL	C	1981	O, S
On the performance enhancement of paging systems through program analysis and transformations [88]	IEEE	TCO	J	1981	D, S
Using Peephole Optimization on Intermediate Code [89]	ACM	TOPLAS	J	1982	-
Optimizing delayed branches [90]	ACM	MICRO	W	1982	D
A composite hoisting-strength reduction transformation for global program optimization part i [91]	T&F	JCM	J	1982	E
Optimization of range checking [92]	ACM	PLDI (SCC)	C	1982	E
Register allocation and spilling via graph coloring [93]	ACM	PLDI	C	1982	A, D, E, M, O, S
Experience with the SETL optimizer [94]	ACM	TOPLAS	J	1983	S
Postpass Code Optimization of Pipeline Constraints [95]	ACM	TOPLAS	J	1983	A, E
Conversion of control dependence to data dependence [96]	ACM	POPL	C	1983	O, S
Code selection through object code optimization [97]	ACM	TOPLAS	J	1984	-
Register allocation and exhaustive peephole optimization [98]	Wiley	SPE	J	1984	E
Automatic generation of peephole optimizations [99]	Springer	CC	C	1984	E
Analysis of interprocedural side effects in a parallel programming environment [100]	Springer	ICSP	C	1984	O
Polyvariant mixed computation for analyzer programs [101]	Springer	Acta Inf.	J	1984	O
Stream processing [102]	ACM	LFP	C	1984	O
A hierarchical basis for reordering transformations [103]	ACM	POPL	C	1984	O
Register allocation by priority-based coloring [104]	ACM	PLDI (SCC)	C	1984	A, E, O
Automatic loop interchange [105]	ACM	PLDI (SCC)	C	1984	O, S
Efficient computation of flow insensitive interprocedural summary information [106]	ACM	PLDI (SCC)	C	1984	A, O
Program transformations in a denotational setting [107]	ACM	TOPLAS	J	1985	-
On linearizing parallel code [108]	ACM	POPL	C	1985	O
Distributed execution of functional programs using serial combinators [109]	IEEE	TCO	J	1985	S
Strictness analysis-a practical approach [110]	Springer	FPCA	C	1985	S
A linear algorithm for finding dominators in flow graphs and related problems [111]	ACM	STOC	C	1985	E, M, O
Advanced compiler optimizations for supercomputers [112]	ACM	CACM	J	1986	-
Efficient compilation of linear recursive functions into object level loops [113]	ACM	PLDI (SCC)	C	1986	-
The impact of interprocedural analysis and optimization in the Rn programming environment [114]	ACM	TOPLAS (LOPES)	C	1986	-
Efficient instruction scheduling for a pipelined architecture [115]	ACM	PLDI (SCC)	C	1986	A
Efficient symbolic analysis of programs [116]	ACM	JCSS	J	1986	A
Graph-Based Algorithms for Boolean Function Manipulation [117]	IEEE	TC	J	1986	D
Loops skewing: The wavefront method revisited [118]	Springer	JPP	J	1986	O
Highly concurrent scalar processing [119]	ACM	CAN	J	1986	O
Multiplication by integer constants [120]	Wiley	SPE	J	1986	S
Global register allocation at link time [121]	ACM	PLDI	C	1986	A, E

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Title	Venue Type	Venue Name	Type	Year	Sources
Interprocedural constant propagation [122]	ACM	PLDI	C	1986	A, D, E, O, S
Interprocedural optimization: eliminating unnecessary recompilation [123]	ACM	CC	C	1986	A, E, O
Interprocedural dependence analysis and parallelization [124]	ACM	PLDI	C	1986	O, S
Effectiveness of a machine-level, global optimizer [125],	ACM	PLDI (SCC)	C	1986	A, E
Direct parallelization of call statements [126]	ACM	PLDI (SCC)	C	1986	O, S
Code motion of control structures in high-level languages [127]	ACM	POPL	C	1986	E, M
Automatic inference and fast interpretation of peephole optimization rules [128]	Wiley	SPE	J	1987	E
Compiler Algorithms for Synchronization [129]	IEEE	TCO	J	1987	O
Automatic decomposition of scientific programs for parallel execution [130]	ACM	POPL	C	1987	O
Guided Self-Scheduling: A Practical Scheduling Scheme for Parallel Supercomputers [131]	IEEE	TCO	J	1987	S
Loop coalescing: A compiler transformation for parallel machines [132]	ACM	ICPP	C	1987	S
Strategies for cache and local memory management by global program transformation [133]	Springer	ICS	C	1987	S
The program dependence graph and its use in optimization [134]	ACM	TOPLAS	J	1987	A, E, M, O
Automatic translation of Fortran programs to vector form [135]	ACM	TOPLAS	J	1987	D, O, S
Safety consideration for storage allocation optimizations [136]	ACM	PLDI	C	1988	-
Efficient and correct execution of parallel programs that share memory [137]	ACM	TOPLAS (LOPES)	C	1988	-
On the control dependence in the program dependence graph [138]	ACM	CSC	C	1988	-
Resource requirements of dataflow programs [139]	ACM	ISCA	C	1988	-
Incremental data flow analysis via dominator and attribute update [140]	ACM	POPL	C	1988	-
An efficient approach to data flow analysis in a multiple pass global optimizer [141]	ACM	PLDI	C	1988	A
A solution to a problem with Morel and Renvoise's "Global optimization by suppression of partial redundancies" [142]	ACM	TOPLAS	J	1988	E
Efficient computation of flow-insensitive interprocedural summary information—a correction [143]	ACM	SIGPLAN Notices	J	1988	O
Dependence of multi-dimensional array references [144]	ACM	ICS	C	1988	O
The importance of direct dependences for automatic parallelization [145]	ACM	ICS	C	1988	O
Introducing symbolic problem solving techniques in the dependence testing phases of a vectorizer [146]	ACM	ICS	C	1988	O
Generating sequential code from parallel code [147]	ACM	SC	C	1988	O
CRegs: a new kind of memory for referencing arrays and pointers [148]	ACM	ICS	C	1988	-
Advanced loop optimizations for parallel computers [149]	ACM	ICS	C	1988	S
An introduction to a formal theory of dependence analysis [150]	Springer	JSC	J	1988	S
Analysis of interprocedural side effects in a parallel programming environment [151]	ACM	ICS	C	1988	S
Array expansion [152]	ACM	ICS	C	1988	S
Loop quantization: A generalized loop unwinding technique [153]	ACM	JPDC	J	1988	S
Supernode partitioning [154]	ACM	PLDI	C	1988	S
A fast algorithm for code movement optimisation [155]	ACM	PLDI	C	1988	A, E
Compiling programs for distributed-memory multiprocessors [156]	Elsevier	JSC	J	1988	O, S
Optimal loop parallelization [157]	ACM	PLDI	C	1988	A, E, M, S
Software pipelining: an effective scheduling technique for VLIW machines [158]	ACM	PLDI	C	1988	D, E

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Title	Venue Type	Venue Name	Type	Year	Sources
The program summary graph and flow-sensitive interprocedural data flow analysis [159]	ACM	PLDI	C	1988	A, O
A framework for determining useful parallelism [160]	ACM	ICS	C	1988	O, S
An overview of the PTRAN analysis system for multiprocessing [161]	Elsevier	JPDC	J	1988	D, S
Efficient interprocedural analysis for program parallelization and restructuring [162]	ACM	PPoPP (PPEALS)	C	1988	O, S
Compiling issues for supercomputers [163]	ACM/IEE	SC	C	1988	O, S
Minimizing register usage penalty at procedure calls [164]	ACM	PLDI	C	1988	A, S
Estimating interlock and improving balance for pipelined architectures [165]	Elsevier	JPDC	J	1988	O, S
Global value numbers and redundant computations [166]	ACM	POPL	C	1988	E, O
Perfect pipelining: A new loop parallelization technique [167]	Springer	ESOP	C	1988	A, S
Code scheduling and register allocation in large basic blocks [168]	ACM	SC	C	1988	A, E
Interprocedural side-effect analysis in linear time [169]	ACM	PLDI	C	1988	A, E, O
A technique for summarizing data access and its use in parallelism enhancing transformations [170]	ACM	PLDI	C	1989	-
Interprocedural analysis vs. procedure integration [171]	Elsevier	IPL	J	1989	-
Unified management of registers and cache using liveness and cache bypass [172]	ACM	PLDI	C	1989	A
A new algorithm for composite hoisting and strength reduction optimisation [173]	T&F	JCM	J	1989	E
Interprocedural data flow testing [174]	ACM	SEN	C	1989	-
Program optimization for instruction caches [175]	ACM	ASPLOS	C	1989	A
Dependence analysis for pointer variables [176]	ACM	PLDI	C	1989	E
The program dependence graph and vectorization [177]	ACM	POPL	C	1989	O
Achieving high instruction cache performance with an optimizing compiler [178]	ACM	ISCA	C	1989	S
Evaluating the performance of four snooping cache coherency protocols [179]	IEEE	ICSA	C	1989	S
Scans as primitive parallel operations [180]	IEEE	TCO	J	1989	S
Code generation using tree matching and dynamic programming [181]	ACM	TOPLAS	J	1989	A, D, E, M
Register allocation via clique separators [182]	ACM	PLDI	C	1989	A, E
Fast interprocedural alias analysis [183]	ACM	POPL	C	1989	A, E, O, S
Coloring heuristics for register allocation [184]	ACM	PLDI	C	1989	A, E, O
More iteration space tiling [185]	ACM/IEE	SC	C	1989	A, S
Data dependence analysis on multi-dimensional array references [186]	ACM	ICS	C	1989	O, S
Spill code minimization techniques for optimizing compilers [187]	ACM	PLDI	C	1989	A, E
Customization: Optimizing compiler technology for SELF, a dynamically-typed OOP language [188]	ACM	PLDI	C	1989	O, S
An efficient method of computing static single assignment form [189]	ACM	POPL	C	1989	A, O
The value flow graph: A program representation for optimal program transformations [190]	Springer	ESOP	C	1990	-
Compiler techniques for data partitioning of sequentially iterated parallel loops [191]	ACM	SC	C	1990	-
Vectorization of tree traversals [192]	Elsevier	JCP	J	1990	-
Loop displacement: an approach for transforming and scheduling loops for parallel execution [193]	ACM	SC	C	1990	-
Experience with interprocedural analysis of array side effects [194]	ACM	SC	C	1990	-
An approach to ordering optimizing transformations [195]	ACM	PPoPP	C	1990	A
Register allocation across procedure and module boundaries [196]	ACM	PLDI	C	1990	A
Region Scheduling: An Approach for Detecting and Redistributing Parallelism [197]	IEEE	TSE	J	1990	E

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Title	Venue Type	Venue Name	Type	Year	Sources
Constructing the procedure call multigraph [198]	IEEE	TSE	J	1990	O
On the perfect accuracy of an approximate subscript analysis test [199]	ACM	CAN	J	1990	O
Structured dataflow analysis for arrays and its use in an optimizing compiler [200]	Wiley	Software: Practice and Experience	J	1990	O
Compilation of Haskell array comprehensions for scientific computing [201]	ACM	PLDI	C	1990	S
How to read floating point numbers accurately [202]	ACM	PLDI	C	1990	S
How to print floating-point numbers accurately [203]	ACM	PLDI	C	1990	S
Profile guided code positioning [204]	ACM	PLDI	C	1990	S
Updating distributed variables in local computations [205]	Wiley	SPE	J	1990	S
An interval-based approach to exhaustive and incremental interprocedural data-flow analysis [206]	ACM	TOPLAS	J	1990	E, O
The priority-based coloring approach to register allocation [207]	ACM	TOPLAS	J	1990	A, D, E, S
Improving register allocation for subscripted variables [208]	ACM	PLDI	C	1990	A, E, M, O, S
Analysis of pointers and structures [209]	ACM	PLDI	C	1990	A, E
Loop distribution with arbitrary control flow [210]	ACM	PLDI	C	1990	O, S
Graph coloring register allocation for processors with multi-register operands [211]	ACM	PLDI	C	1990	A, E
A data locality optimizing algorithm [212]	ACM	PLDI	C	1991	-
On optimal parallelization of arbitrary loops [213]	Elsevier	JPDC	J	1991	-
On the adequacy of dependence-based representations for programs with heaps [214]	Springer	TACS	C	1991	-
Automatic construction of sparse data flow evaluation graphs [215]	ACM	POPL	C	1991	-
Compiler optimizations for Fortran D on MIMD distributed-memory machines [216]	ACM	SC	C	1991	-
Register allocation via hierarchical graph coloring [217]	ACM	PLDI	C	1991	A
Circular scheduling: a new technique to perform software pipelining [218]	ACM	PLDI	C	1991	A
Efficient DAG construction and heuristic calculation for instruction scheduling [219]	ACM	MICRO	W	1991	E
Efficiently computing static single assignment form and the control dependence graph [220]	ACM	TOPLAS	J	1991	M
Software prefetching [221]	ACM	ASPLOS	C	1991	O
Compiling global name-space parallel loops for distributed execution [222]	IEEE	TPDS	J	1991	O
An implementation of interprocedural bounded regular section analysis [223]	ACM	TPDS	J	1991	O
Limits of instruction-level parallelism [224]	ACM	ASPLOS	C	1991	S
Uniform techniques for loop optimization [225]	ACM	ICS	C	1991	S
A data locality optimizing [212]	ACM	PLDI	C	1991	A, D, M, O
Constant propagation with conditional branches [226]	ACM	TOPLAS	J	1991	A, E, M, O, S
Efficient and exact data dependence analysis [227]	ACM	PLDI	C	1991	A, D, O
Efficiently computing static single assignment [228]	ACM	TOPLAS	J	1991	A, D, E, O
Global instruction scheduling for superscalar machines [229]	ACM	PLDI	C	1991	A, D, E
Practical adaption of the global optimization algorithm of Morel and Renvoise [230]	ACM	TOPLAS	J	1991	A, E
The cache performance and optimizations of blocked algorithms [231]	ACM	ASPLOS	C	1991	A, D
A loop transformation theory and an algorithm to maximize parallelism [232]	IEEE	TPDS	J	1991	O, S

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Title	Venue Type	Venue Name	Type	Year	Sources
Dataflow analysis of array and scalar references [233]	Springer	JPP	J	1991	A, S
Optimization of array accesses by collective loop transformations [234]	ACM	SC	C	1991	D, O
Interprocedural transformations for parallel code generation [235]	ACM/IEE	SC	C	1991	O, S
Practical dependence testing [236]	ACM	PLDI	C	1991	A, O
Procedure merging with instruction caches [237]	ACM	PLDI	C	1991	A, S
An experiment with inline substitution [238]	Wiley	SPE	J	1991	E, S
Access normalization: loop restructuring for NUMA compilers [?]]	ACM	ASPLOS	C	1991	-
An efficient resource-constrained global scheduling technique for superscalar and VLIW processors [239]	ACM	MICRO	W	1992	-
Non-unimodular transformations of nested loops [240]	ACM	SC	C	1992	-
A general framework for iteration-reordering loop transformations [241]	ACM	PLDI	C	1992	-
A program integration algorithm that accommodates semantics-preserving transformations [242]	ACM	TOSEM	J	1992	-
The transitive closure of control dependence: The iterated join [243]	ACM	TOPLAS (LOPLAS)	J	1992	-
Abstract description of pointer data structures: an approach for improving the analysis and optimization of imperative programs [244]	ACM	TOPLAS (LOPLAS)	J	1992	-
Abstractions for recursive pointer data structures: improving the analysis and transformation of imperative programs [245]	ACM	PLDI	C	1992	-
Generalized dominators and post-dominators [246]	ACM	POPL	C	1992	-
A comprehensive approach to parallel data flow analysis [247]	ACM	ICS	C	1992	-
Integrating scalar optimization and parallelization [248]	Springer	LCPC	C	1992	A
Sharlit—a tool for building optimizers [249]	ACM	PLDI	C	1992	A
How to analyze large programs efficiently and informatively [250]	ACM	PLDI	C	1992	A
Compiler code transformations for superscalar-based high performance systems [251]	IEEE	SC	C	1992	A
Engineering a simple, efficient code-generator generator [252]	ACM	TOPLAS (LOPLAS)	J	1992	D
Some efficient solutions to the affine scheduling problem. I. One-dimensional time [253]	Elsevier	JPP	J	1992	D
Avoiding unconditional jumps by code replication [254]	ACM	PLDI	C	1992	E
Rematerialization [255]	ACM	PLDI	C	1992	E
Coloring register pairs [256]	ACM	TOPLAS (LOPLAS)	J	1992	E
Optimizing for parallelism and data locality [257]	ACM	SC	C	1992	O
Maximizing loop parallelism and improving data locality via loop fusion and distribution [258]	ACM	LCPC	W	1992	O
Beyond induction variables [259]	ACM	PLDI	C	1992	O
Efficient call graph analysis [260]	ACM	TOPLAS (LOPLAS)	J	1992	O
A safe approximate algorithm for interprocedural aliasing [261]	ACM	PLDI	C	1992	O
Relaxing SIMD control flow constraints using loop transformations [245]	ACM	PLDI	C	1992	S
Unexpected side effects of inline substitution: a case study [262]	ACM	TOPLAS (LOPLAS)	J	1992	S
Eliminating false positives using the omega test [263]	ACM	PLDI	C	1992	A, D, O
Lazy code motion [264]	ACM	PLDI	C	1992	A, D, E
Software support for speculative loads [265]	ACM	ASPLOS	C	1992	A, E
A practical algorithm for exact array dependence analysis [266]	ACM	CACM	J	1992	O, S
Array privatization for parallel execution of loops [267]	ACM	ICS	C	1992	O, S
The power test for data dependence [268]	IEEE	TCO	J	1992	A, S

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Title	Venue Type	Venue Name	Type	Year	Sources
Eliminating branches using a superoptimizer and the GNU C compiler [269]	ACM	PLDI	C	1992	E, S
Design and evaluation of a compiler algorithm for prefetching [270]	ACM	ASPLOS	C	1992	D, O
Data dependence and data-flow analysis of arrays [271]	Springer	LCPC	C	1992	-
Advanced compiler optimizations for sparse computations [272]	ACM	SC	C	1993	-
Access normalization: loop restructuring for NUMA computers [273]	ACM	TOCS	J	1993	-
On the conversion of indirect to direct recursion [274]	ACM	TOPLAS (LOPLAS)	J	1993	-
A practical data flow framework for array reference analysis and its use in optimizations [275]	ACM	PLDI	C	1993	-
Using lifetime predictors to improve memory allocation performance [276]	ACM	PLDI	C	1993	-
Data flow analysis for parallel programs [277]	ACM	CSC	C	1993	-
Dependence-based program analysis [278]	ACM	PLDI	C	1993	A
A variation of Knoop, Rüthing, and Steffen’s lazy code motion [279]	ACM	PLDI	C	1993	E
Lazy strength reduction [280]	C&H	JPL	J	1993	A, E
Collective loop fusion for array contraction [281]	Springer	LCPC	C	1993	O
Interprocedural constant propagation: an empirical study [282]	ACM	TOPLAS (LOPLAS)	J	1993	O
Symbolic analysis: A basis for parallelization, optimization, and scheduling of programs [283]	Springer	LCPC	W	1993	O
Array-data flow analysis and its use in array privatization [284]	ACM	POPL	C	1993	S
Automatic array alignment in data-parallel programs [285]	ACM	POPL	C	1993	S
Global optimizations for parallelism and locality on scalable parallel machines [286]	ACM	PLDI	C	1993	S
Instruction-Level Parallel Processing: History, Overview, and Perspective [287]	Springer	JSC	J	1993	S
A practical system for intermodule code optimization at link-time [288]	C&H	JPL	J	1993	A
Loop-level parallelism in numeric and symbolic programs [289]	IEEE	TPDS	J	1993	S
Orchestrating interactions among parallel computations [290]	ACM	PLDI	C	1993	S
The superblock: an effective technique for VLIW and superscalar compilation [291]	Springer	JSC	J	1993	S
Optimizing array bound checks using flow analysis [292]	ACM	PLDI	C	1993	A, E
Register allocation with instruction scheduling [293]	ACM	PLDI	C	1993	A, E
A methodology for procedure cloning [294]	Elsevier	COLA (COMLAN)	J	1993	A, O, S
Efficient flow-sensitive interprocedural computation of pointer-induced aliases and side effects [295]	ACM	POPL	C	1993	E, O, S
Automatic array privatization [296]	Springer	LCPC	W	1993	O, S
Interprocedural modification side effect analysis with pointer aliasing [297]	ACM	PLDI	C	1993	M, S
Branch prediction for free [298]	ACM	PLDI	C	1993	A, M
Enabling unimodular transformations [299]	ACM	SC	C	1994	-
Compiler optimizations for improving data locality [300]	ACM	ASPLOS	C	1994	-
The revival transformation [301]	ACM	PLDI	C	1994	-
The range test: a dependence test for symbolic, non-linear expressions [302]	ACM	SC	C	1994	-
Link-time optimization of address calculation on a 64-bit architecture [303]	ACM	PLDI	C	1994	A
Effective partial redundancy elimination [304]	ACM	PLDI	C	1994	A
Partial dead code elimination [305]	ACM	PLDI	C	1994	A
A general data dependence test for dynamic, pointer-based data structures [244]	ACM	PLDI	C	1994	A

Continued on next page

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Title	Venue Type	Venue Name	Type	Year	Sources
Instruction scheduling over regions: A framework for scheduling across basic blocks [306]	Springer	CC	C	1994	A
Value dependence graphs: representation without taxation [307]	ACM	POPL	C	1994	A
Zero-cost range splitting [308]	ACM	PLDI	C	1994	E
Reducing branch costs via branch alignment [309]	ACM	OSR	J	1994	E
Improving the ratio of memory operations to floating-point operations in loops [310]	ACM	TOPLAS	J	1994	M
Optimizing multi-method dispatch using compressed dispatch tables [311]	ACM	OOPSLA	C	1994	M
Improving the accuracy of static branch prediction using branch correlation [312]	ACM	ASPLOS	C	1994	M
A compiler framework for restructuring data declarations to enhance cache and TLB effectiveness [313]	IEEE	CASCON	C	1994	S
False sharing and spatial locality in multiprocessor caches [314]	IEEE	TCO	J	1994	S
Reassociation and strength reduction [315]	ACM	SCO	C	1994	S
The alignment-distribution graph [316]	Springer	LCPC	W	1994	S
Improvements to graph coloring register allocation [317]	ACM	TOPLAS	J	1994	A, E, M
Context-sensitive interprocedural points-to analysis in the presence of function pointers [318]	ACM	PLDI	C	1994	D, E
Interprocedural may-alias analysis for pointers: beyond k-limiting [319]	ACM	PLDI	C	1994	A, E
Scalar replacement in the presence of conditional control flow [320]	Wiley	SPE	J	1994	E, O

Table 1: The papers from *Advanced Compiler Design and Implementation* [1] (A), *Compilers: Principles, Techniques, and Tools* [2] (D), *Engineering a Compiler* [3] (E), *Modern Compiler Implementation in C/Java/ML* [4–6] (M), *Optimizing Compilers for Modern Architectures: A Dependence-Based Approach* [7] (O), and the Bacon’s survey [8] (S). In the **Type** column, C stands for conference, J for journal, and W for workshop.

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